

Each question is worth 5 points for a total of 30 points. The extra Credit can get you an additional 5 points making the highest possible score 35 out of 30. Write solutions using COMPLETE sentences.

Problem 1

Prove every integer is either even or odd, but not both.

Problem 2

Let a , b , and $n \in \mathbb{Z}$ with $n > 0$. Prove $a \equiv b \pmod{n} \iff a \bmod n = b \bmod n$.

Problem 3

Find x and y such that $14489x + 98y = 1$.

Problem 4

Solve the system of congruence relations: $x \equiv 3 \pmod{7}$, $x \equiv 0 \pmod{4}$, and $x \equiv 8 \pmod{25}$.

Problem 5

Suppose a , b , and $p \in \mathbb{Z}$ where p is a prime. If $p|ab$, then $p|a$ or $p|b$.

Problem 6

Suppose $p, q_1, q_2, \dots, q_t$ are prime numbers. Prove that if $p|(q_1 \cdot q_2 \cdot \dots \cdot q_t)$ then $p = q_i$ for $1 \leq i \leq t$.

Extra Credit:

Prove $\sqrt{2}$ is irrational.